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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

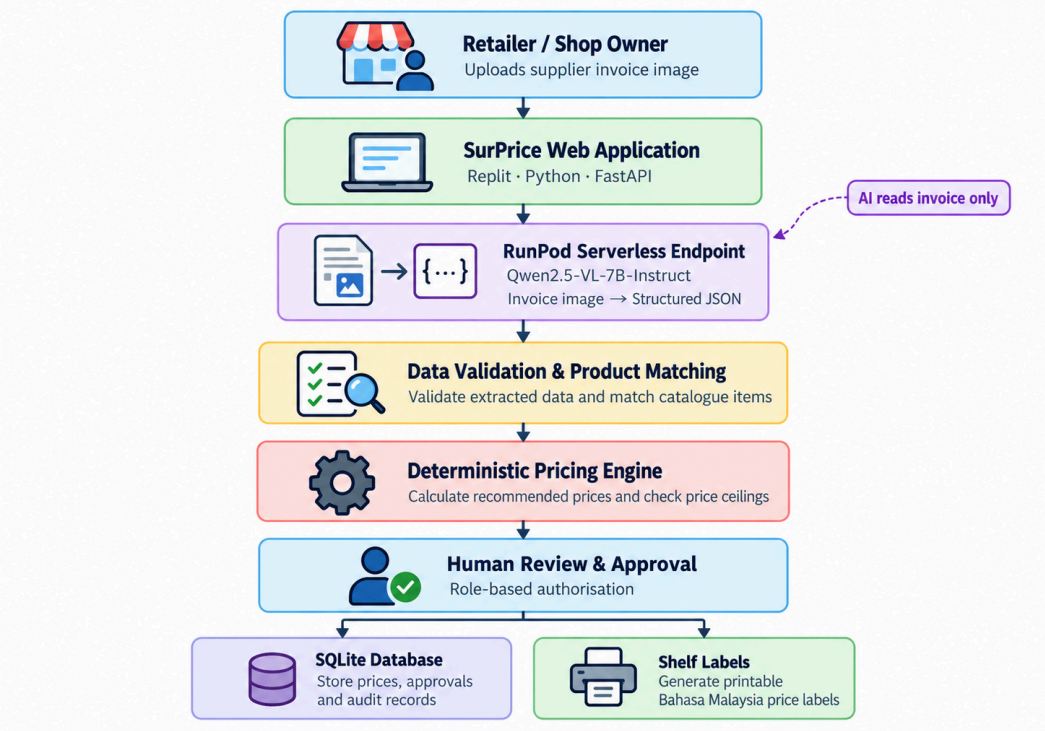
Team Name	Umai
Project / Solution Name	SurPrice
Name (Leader)	Ahmad Zhafir Bin Razali
Email & Phone (Leader)	zhafirrazali@gmail.com | +601112946874
University Affiliation(s)	Universiti Putra Malaysia
Live Prototype URL	https://surprice.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/WanHafizuddin/SURPrice

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]



SurPrice is a cloud-deployed retail pricing application developed using Python, FastAPI, Jinja2, and SQLite, with Replit providing the application hosting environment and RunPod Serverless providing GPU-powered AI inference.

Retailers access the application through a web browser to upload supplier invoice images. The FastAPI backend sends invoice images to a RunPod Serverless Endpoint running the Qwen2.5-VL-7B-Instruct vision-language model. The model extracts supplier information, product names, quantities, units, and costs into structured JSON.

The backend validates the extracted information and matches invoice items against the retailer's existing product catalogue. A deterministic pricing engine calculates recommended selling prices based on target profit margins, flags significant cost changes, and checks applicable price-control rules.

Proposed changes undergo role-based human approval. Approved prices are recorded in SQLite alongside audit records, and printable Bahasa Malaysia shelf labels are generated.

The system separates AI-powered document interpretation from pricing decisions to maintain transparency, reliability, and human oversight. If AI reading fails or times out, staff enter the invoice lines manually; a clearly labelled Simulated AI demo mode runs only when RunPod is not configured.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.11+	Application programming language	PSF License
FastAPI 0.161.1	Backend web framework	MIT
Uvicorn 0.54.0	ASGI application server	BSD-3-Clause
Jinja2 3.1.6	HTML template rendering	BSD-3-Clause
python-multipart 0.0.32	Invoice image and form uploads	Apache-2.0
itsdangerous 2.2.0	Session signing support	BSD-3-Clause
httpx 0.28.1	HTTP communication with RunPod API	BSD-3-Clause
Pillow 13.0.0	Invoice image processing	HPND
SQLite	Local relational database	Public domain
Starlette	Application middleware and sessions	BSD-3-Clause
PyTorch	Model inference and GPU computation	BSD-3-Clause
Transformers	Loads and executes Qwen2.5-VL	Apache-2.0
Accelerate	Supports model device management	Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Serverless REST API (RunPod)	Sends supplier invoice images to the deployed Qwen2.5-VL-7B-Instruct model and retrieves structured extraction results	RunPod endpoint (sponsor-provided)

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Hosting and deployment of the FastAPI web application	Replit Core (sponsor-provided)
RunPod Serverless	GPU inference for Qwen2.5-VL-7B-Instruct through a serverless endpoint	Serverless GPU (sponsor-provided)

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen2.5-VL-7B-Instruct	Qwen / Alibaba Cloud, hosted on RunPod	Pretrained vision-language model used for supplier invoice extraction	Apache-2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic Invoice Dataset	Repository: fixtures/invoices/	Synthetic (team-created)	Team-owned	No
Pricing Rule-Breach Test Cases	Repository: fixtures/breaches.json	Synthetic (team-created)	Team-owned	No

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code	Code generation, implementation assistance, debugging, and development	Major
ChatGPT	Proposal drafting, documentation review, technical descriptions	Moderate
OpenAI image generation	Generating conceptual architecture diagrams	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team used AI-assisted development tools to support coding, documentation, and design while retaining responsibility for the final project.

Generated code was reviewed against application requirements and supported by automated tests covering pricing calculations, invoice validation, authentication, approval workflows, and database operations.

Synthetic test cases were used to assess pricing safeguards. The RunPod inference integration was also evaluated using a synthetic invoice, with all eight product lines reportedly extracted correctly.

AI-generated documentation and diagrams were checked against the implemented architecture. Further testing with diverse real-world invoices is required to assess extraction accuracy and reliability.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Cloud development and hosting platform	Used to host and deploy the SurPrice FastAPI prototype
RunPod	Serverless GPU inference platform / sponsored compute credits	Hosts the Qwen2.5-VL-7B-Instruct model for AI-powered invoice extraction

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit provides the cloud hosting environment for SurPrice, enabling users and judges to access the deployed web application. RunPod Serverless provides GPU-powered inference for the Qwen2.5-VL-7B-Instruct model, enabling automated invoice extraction from uploaded images.

Both platforms support the current prototype architecture. However, the application uses standard Python and FastAPI technologies, while its invoice extraction component is modular. Therefore, Replit could be replaced with another compatible hosting platform, and RunPod could be replaced with another inference provider.

Deterministic pricing calculations, approval workflows, and database operations remain independent of the AI inference service.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The SurPrice prototype uses fictional retail accounts, products, supplier invoices, and pricing information for demonstration purposes. No real customer or supplier personal data is intentionally included in the synthetic datasets.

The application stores account information, password hashes, invoices, product records, and audit logs in SQLite. Security measures include password hashing, session-based authentication, role-based access controls, CSRF protection, and shop-level data isolation.

Uploaded invoices are stored within the application's private data directory. When RunPod inference is enabled, invoice images are transmitted through HTTPS to the team's RunPod Serverless Endpoint for AI extraction. Pricing calculations and approval decisions remain within the FastAPI application.

API credentials are managed through environment variables rather than exposed in the source code.

A formal data-retention period has not yet been implemented. Future use with real business information will require defined retention policies, appropriate data deletion procedures, and additional privacy safeguards.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

SurPrice integrates live AI-powered invoice extraction through a RunPod Serverless Endpoint. However, vision-language models may misinterpret unclear text, unusual invoice layouts, mixed languages, product descriptions, or numerical values.

Initial evaluation reported correct extraction of all eight lines from one synthetic invoice. This limited test does not establish accuracy across real-world supplier invoices.

RunPod inference also introduces potential GPU cold-start delays, network failures, processing timeouts, and external-service dependency.

To mitigate these risks, SurPrice validates extracted data, flags uncertain product matches, and falls back to manual entry when AI reading fails or times out.

Pricing calculations remain deterministic and require human approval before catalogue updates. The system applies available price-control safeguards and blocks controlled-item changes when valid ceiling information is unavailable.

Current demonstration price-control data is fictional rather than an officially verified dataset. Further work includes testing diverse real invoices, improving inference reliability, and integrating verified pricing references.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



Ahmad Zhafir
24/10/26